THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET B

Chapters 3 & 4: **A Story of Numbers, Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The value of 43 in base 5, in the decimal system, is: 1

(a) 25 **(b)** 23 **(c)** 24 **(d)** 22

2. The mathematician who fully explained computation with the Indian ten-symbol system around 499 CE was: 1

(a) Fibonacci **(b)** Al-Kindi **(c)** Euclid **(d)** Aryabhata

3. The Maya number system was based on: 1

(a) 20 **(b)** 12 **(c)** 10 **(d)** 60

4. The base of the binary number system is: 1

(a) 10 **(b)** 8 **(c)** 16 **(d)** 2

5. Ancient Indian texts named powers of ten such as eka, dasha, shata and: 1

(a) crore only **(b)** lakh only **(c)** sahasra **(d)** yojana

6. Opposite sides of a parallelogram are: 1

(a) only equal **(b)** perpendicular
(c) parallel and equal **(d)** only parallel

7. Every rectangle is a: 1

(a) parallelogram **(b)** square **(c)** kite **(d)** rhombus

8. A rhombus is a parallelogram in which: 1

(a) all angles are 90° **(b)** all sides are equal
(c) one pair is parallel **(d)** diagonals are equal

9. A rectangle that is also a rhombus is a: 1

(a) square **(b)** parallelogram only
(c) trapezium **(d)** kite

10. In rhombus ABCD, diagonal AC bisects angle: 1

(a) B only **(b)** D only **(c)** A and C **(d)** none

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Write XLIV in decimal. 2

12. In base 60 (Mesopotamia), how many minutes make an hour? Why is that a relic of base 60? 2

13. Name the special quadrilaterals whose diagonals are perpendicular. 2

14. The diagonals of a rectangle are 26 cm each. Find AC and BD and say if they bisect each other. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Convert 707_8 to decimal. 3

16. Which special quadrilaterals have all of: four equal sides and equal diagonals? 3

17. Compare Roman and Hindu numeral systems on two points. 3

OR

Describe how the Mesopotamians wrote numbers. 3

18. The diagonals of a quadrilateral bisect each other. Which quadrilaterals can it be? Why? 3

OR

In kite ABCD, AB = AD and CB = CD. If $\angle B = \angle D = 90^\circ$ and $\angle A = 70^\circ$, find $\angle C$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Convert 0.5 and 75 to base 2 where possible.
(ii) What do the digits of 1000_2 represent? 5

OR

Case study: A shopkeeper counts apples in groups of 5, 25 and 125. 5
(i) Which base is this?
(ii) Write 163 apples in this base.
(iii) Write the digits.

20. **Case study:** A tile is a rhombus of side 13 cm with a diagonal of 24 cm. 5
(i) Find the other diagonal.
(ii) Find the area.
(iii) How many tiles cover 3 m²?

OR

(i) Find all angles of a quadrilateral whose angles are in ratio 1:2:3:4. 5
(ii) Is it a trapezium? Explain.

END OF QUESTION PAPER — Total: 40 Marks

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